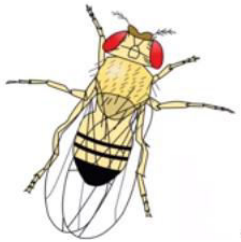
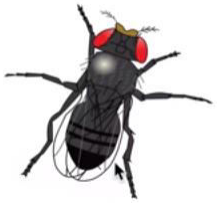
Chapter VI: Linkage dis-equilibrium and Hardy-Weinberg equilibrium

Linkage disequilibrium

Linkage dis-equilibrium

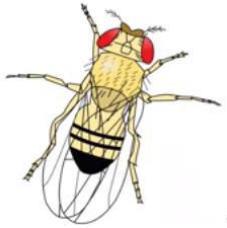


Thomas Morgan
(1866-1945)

P:  x 

Genotype: B B / S S b b / s s

Gametes: B S b s

F1: 100% 

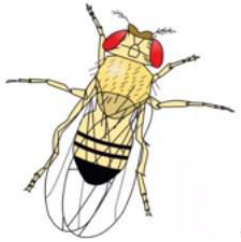
Genotype: B b / S s

Gene recombination

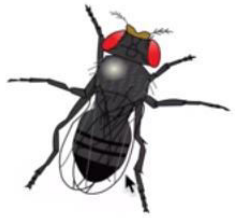


Thomas Morgan
(1866-1945)

BC (Backcross):



x

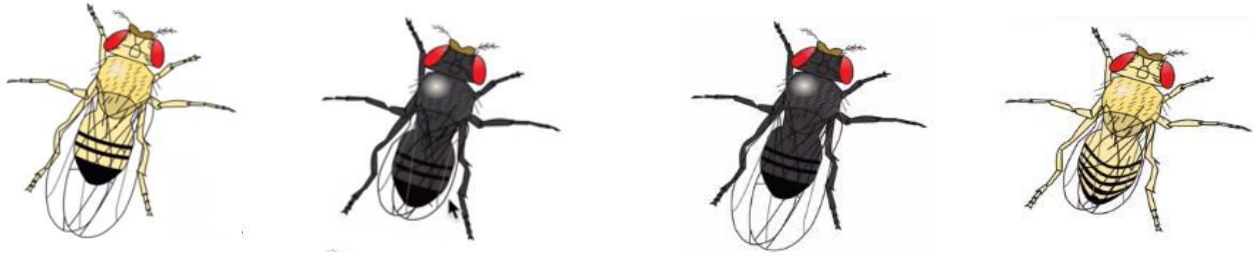


Genotype:

B b
S s

b b
s s

Population BC:



Parental phenotype

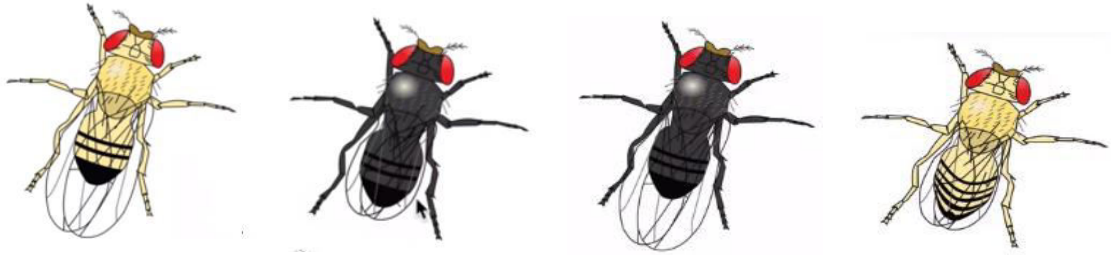
Recombinant phenotype



Proportion: 40% 40% 10% 10%

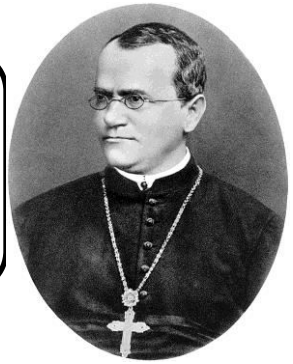
Gene recombination

Population BC:



	B S	b s	b S	B s
b s	B b S s	b b s s	b b S s	B b s s

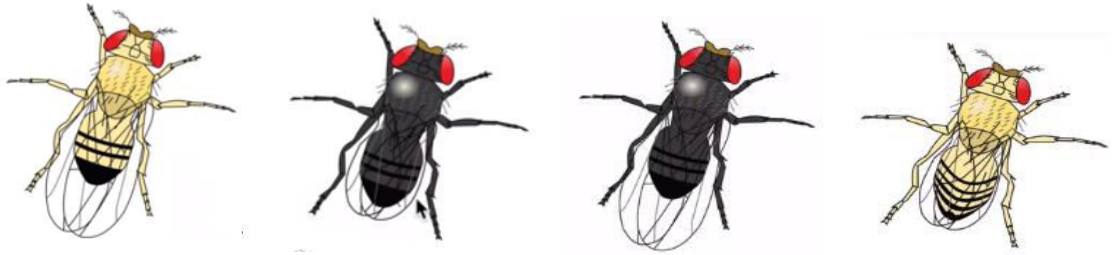
25%	25%	25%	25%



Gregor Mendel

Gene recombination

Population BC:



	B S	b s	b S	B s
b s	B b S s	b b s s	b b S s	B b s s

40% 40%

10% 10%

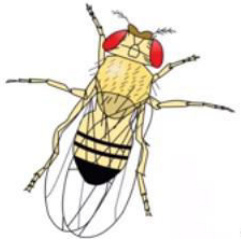
FACT

Gene recombination



Alfred Sturtevant
(1891-1970)

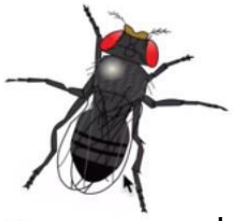
BC (Backcross):



B
S

b
s

X



b
s

b
s

DNA replication:

Meiosis:
Prophase 1

Gametes:

B
S

B
s

b
S

b
s

40%

10%

10%

40%

b
s

100%

Gene recombination

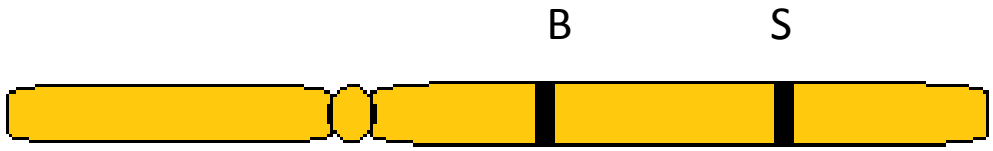


Alfred Sturtevant
(1891-1970)

	40% B S	10% B s	10% b S	40% b s
100% b s	40%	10%	10%	40%



Proportion of recombination between these 2 genes = 10% + 10% = 20%
→ The genetic distance between these 2 genes = 20 cM (**centiMorgan**)



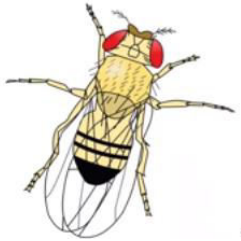
$D_{(B-S)}$ = 20 cM



$D_{(B-R)}$ = 30 cM

Exercises:

BC (Backcross):



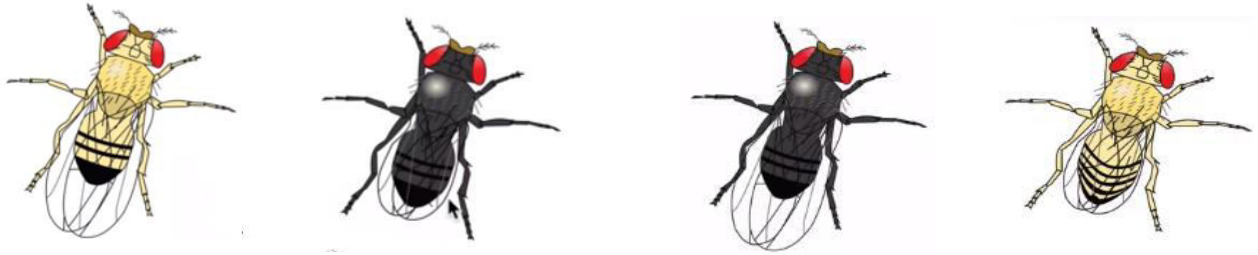
X



Genotype:

$$\frac{B}{S} \,||\, \frac{b}{s} \qquad \frac{b}{s} \,||\, \frac{b}{s}$$

Population BC:



Parental phenotype

Recombinant phenotype



Calculate the proportion of population BC with $D_{(B\text{-}S)}$ = 30 cM

0 cM

Hardy-Weinberg equilibrium

Allele Frequencies in a Population



G.H. Hardy
English
Mathematician



Dr. Wilhelm Weinberg
German Physician

The Hardy-Weinberg equilibrium is a principle stating that the genetic variation in a population will remain constant from one generation to the next in the absence of disturbing factors. When mating is random in a large population with no disruptive circumstances, the law predicts that both genotype and allele frequencies will remain constant because they are in equilibrium.

Conclusions of the Hardy-Weinberg principle

1. Allele frequencies will not change from generation to generation.

2. Genotype proportions determined by the “square law”.

- for two alleles = $(p + q)^2 = p^2 + 2pq + q^2$
- for three alleles $(p + q + r)^2 = p^2 + q^2 + r^2 + 2pq + 2pr + 2qr$

The Equations

$$p + q = 1$$

$$p^2 + 2pq + q^2 = 1$$

- A gene has two alleles, A and a
- The frequency of allele A is represented by p
- The frequency of allele a is represented by q
- The frequency of genotype AA = p^2
- The frequency of genotype aa = q^2
- The frequency of genotype Aa = 2pq

An Example…

Assume a population in which 36% of the population are homozygous for a certain recessive allele, a. Assume the population is at equilibrium.

Question #1: What is the frequency of the recessive allele, a in this population?

$$q^2 = 0.36$$

$$q = \sqrt{0.36}$$

$$q = 0.60$$

An Example…

Assume a population in which 36% of the population are homozygous for a certain recessive allele, a. Assume the population is at equilibrium.

Question #2: What is the frequency of the dominant allele, A in this population?

$$q = 0.60$$

$$p + 0.60 = 1$$

$$p = 0.40$$

An Example…

Assume a population in which 36% of the population are homozygous for a certain recessive allele, a. Assume the population is at equilibrium.

Question #3: What percentage of the population are homozygous for the dominant allele, A?

$$p = 0.40$$

$$p^2 = 0.40^2$$

$$p^2 = 0.16 = 16\%$$

An Example…

Assume a population in which 36% of the population are homozygous for a certain recessive allele, a. Assume the population is at equilibrium.

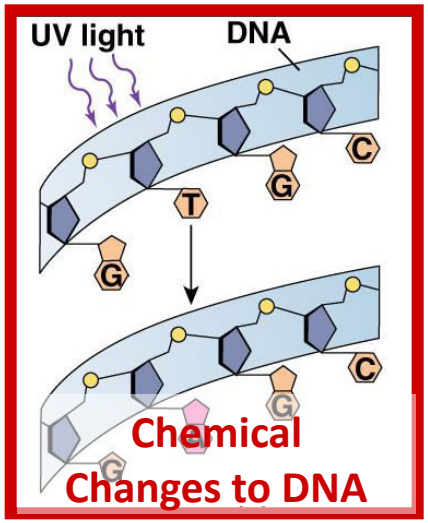
Question #4: What percentage of the population are heterozygous for this trait?

$$2pq = 2(0.40)(0.60)$$

$$2pq = 0.48 = 48\%$$

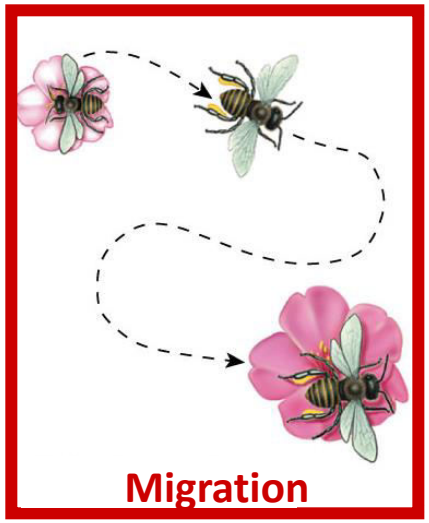
5 Agents of evolutionary change

Mutation



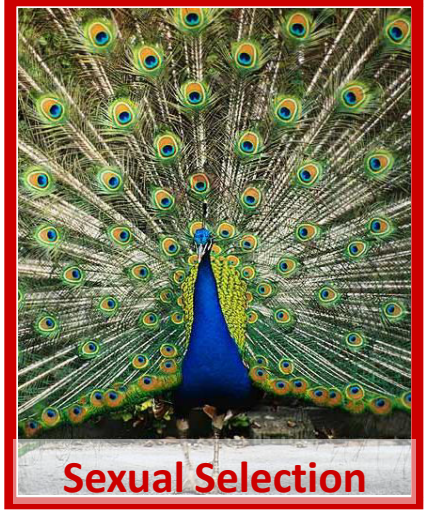
Chemical Changes to DNA

Gene Flow



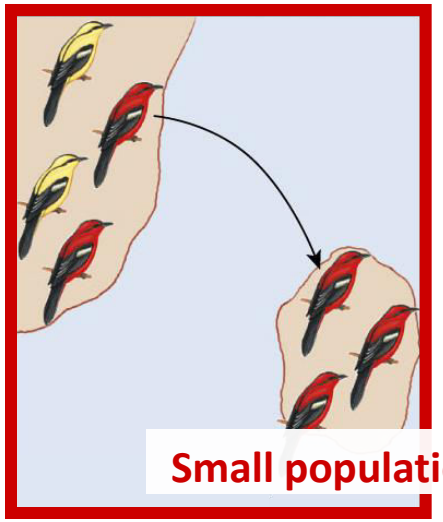
Migration

Non-random mating



Sexual Selection

Genetic Drift



Small population

Selection



Natural Selection
Differential Survival

(e) Selection

Anna Sweetstory has a problem. Her eyes are half-closed all the time. Her teachers think she sleeps during class and her friends think she is always bored. Anna's condition is called ptosis, which is caused by a single dominant gene, *P*. The opposite normal condition, open eyes, is caused by the recessive allele, *p*. Anna's father had ptosis, and his parents were both found to be homozygous dominant with respect to this gene. Anna's mother has normal eyes.

- What is Anna's genotype?
- What are the genotypes of both her parents?
- If Anna marries a man with normal eyes, what is the probability of their having a child with ptosis?

Albinism in humans is inherited as a simple recessive trait. For the following families, what are the genotypes of the parents and offspring?
(Note: When two alternative genotypes are possible, list them both.)

- Two non-albino (normal) parents have five children, four normal and one albino.
- A normal male and an albino female have six children, all normal.